Azure Parts – Loose Discussion

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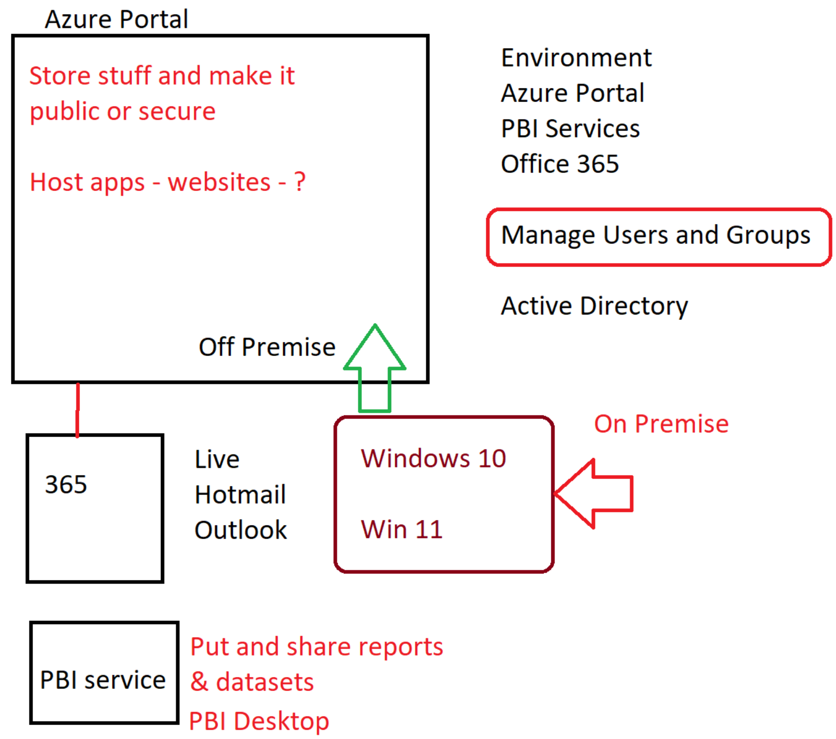
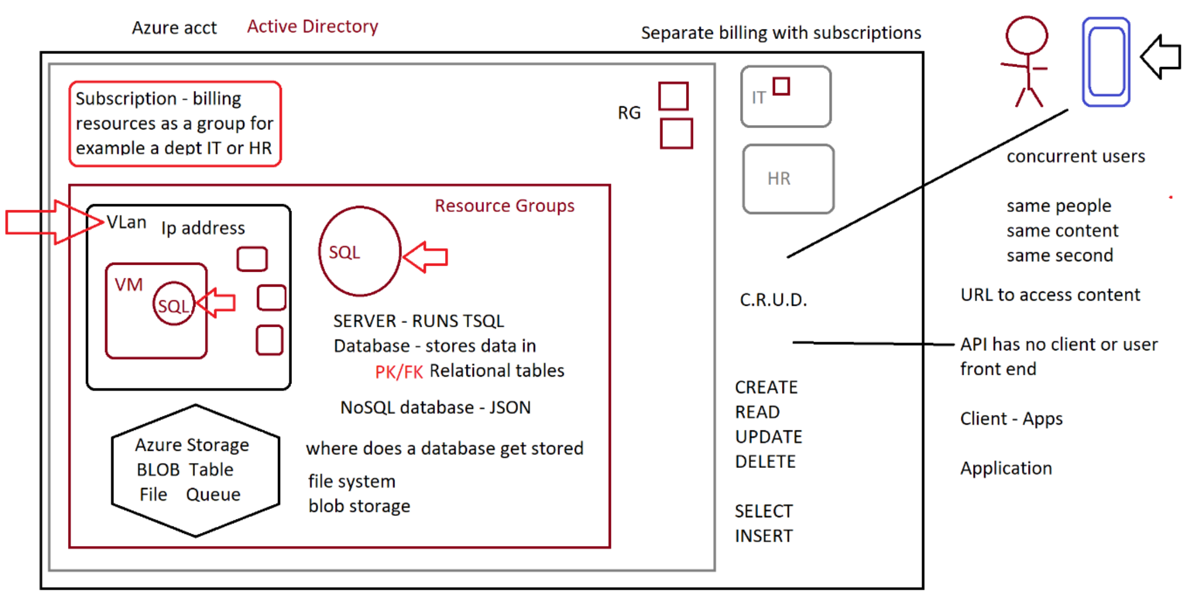
Diagrams


Azure Portal
Store stuff and make it public or secure
Host apps - websites - ?
Off Premise
Environment
Azure Portal
PBI Services
Office 365
Manage Users and Groups
Active Directory
365
Live
Hotmail
Outlook
Windows 10
Win 11
On Premise
PBI service
Put and share reports & datasets
PBI Desktop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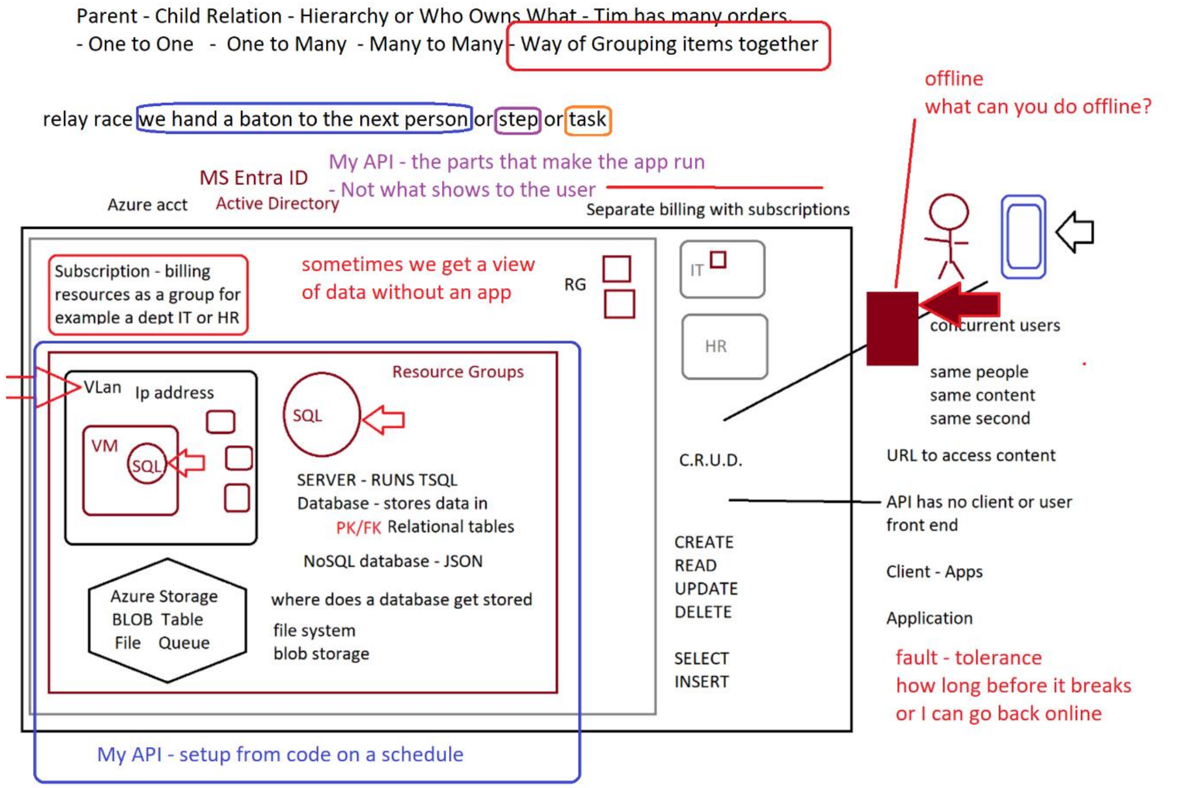
Parent - Child Relation - Hierarchy or Who Owns What - Tim has many orders.
- One to One - One to Many - Many to Many - Way of Grouping items together
relay race we hand a baton to the next person or step or task
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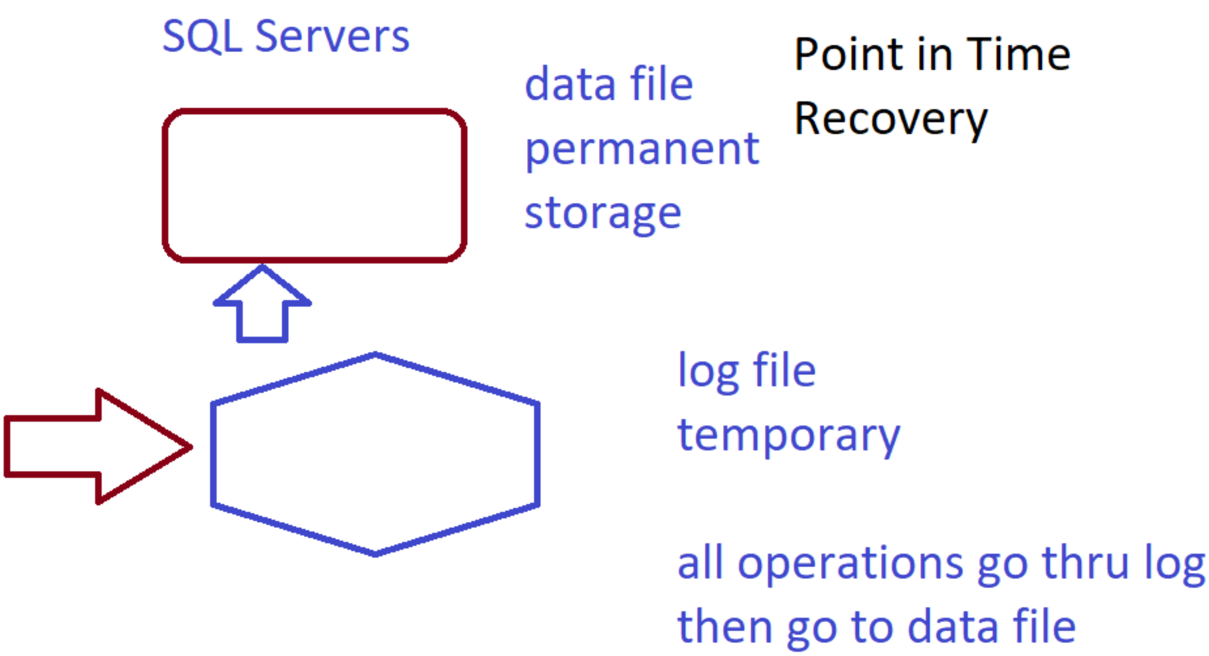
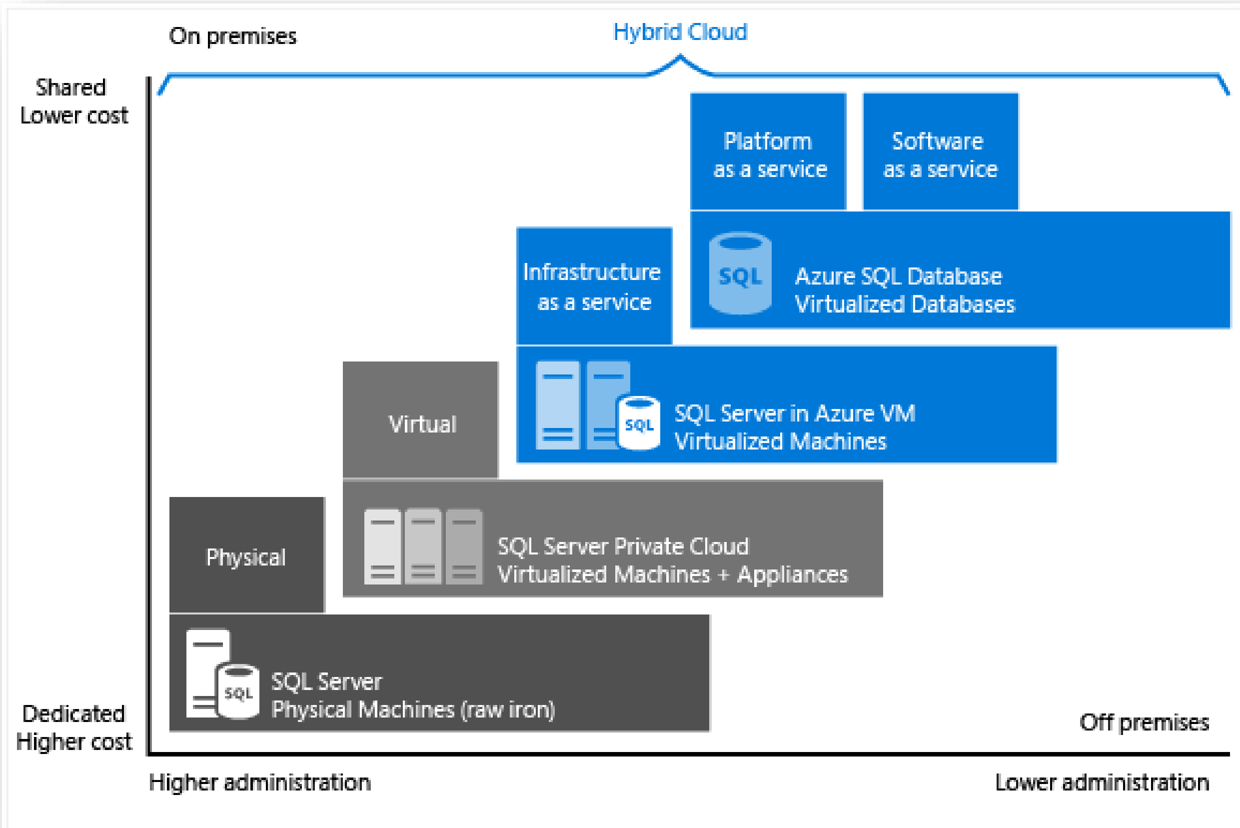
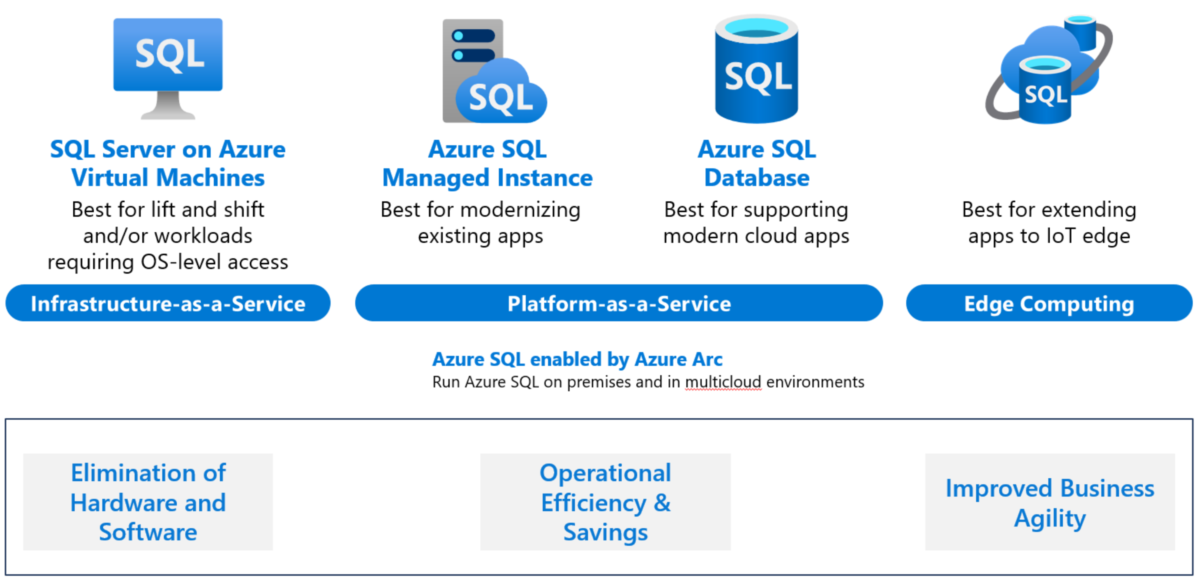

SQL Servers
data file
permanent
storage
Point in Time
Recovery
log file
temporary
all operations go thru log
then go to data file



Azure SQL platform offerings

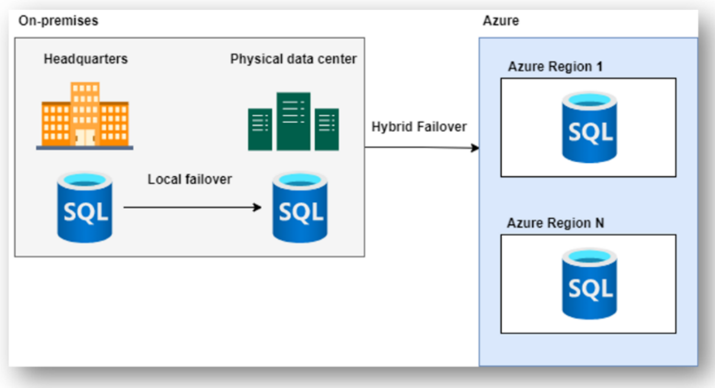


Hybrid scenarios for SQL Server – Disaster Recovery

Most common scenario for a hybrid deployment of SQL Server.

Organizations ensure business continuity during catastrophic events.

Azure is used for DR failover (to one or more regions) while the regular day-to-day processing continues to use on-premises servers for local high availability.

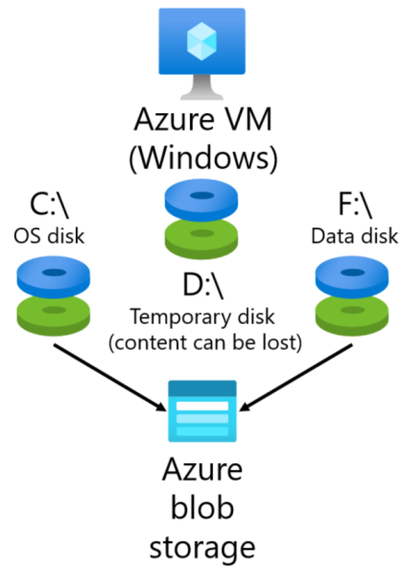


Azure Virtual Machine storage

Each Azure VM has two or more disks:
- OS disk
- Temporary disk
- Data disks (optional)

OS and data disks reside in Azure-based storage service:
- Standard (HDD, SSD)
- Premium (SSD)
- Ultra (SSD)

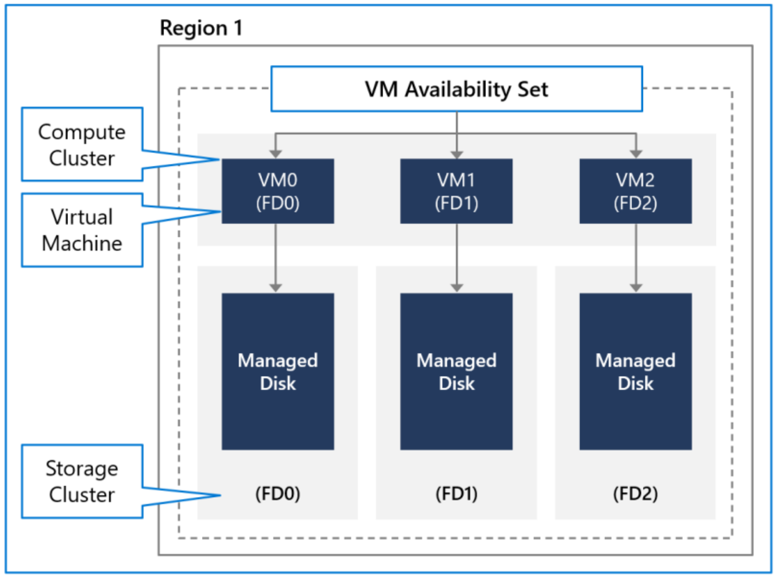
When creating an Azure VM, you can choose between:
- Managed disks (recommended)
- Unmanaged disks



Availability sets

Provide increased availability for multi-VM deployments like Always On Availability Groups, by ensuring the VM are deployed to different physical hosts

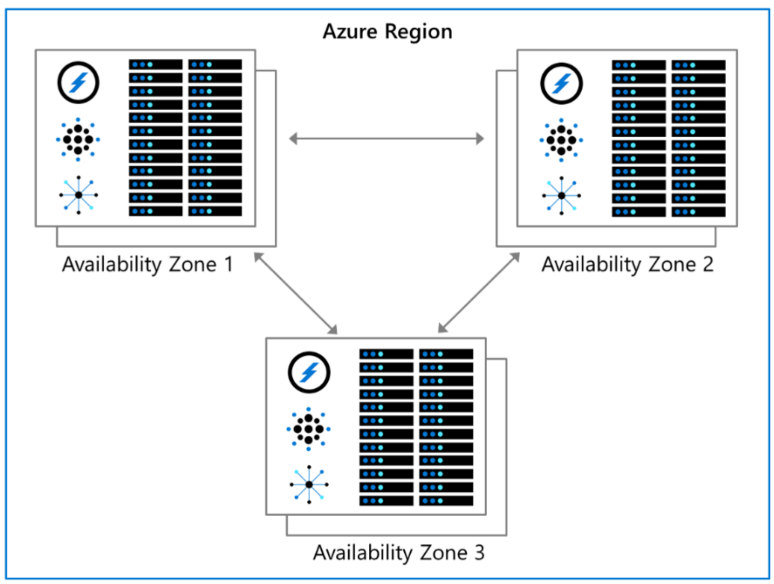
Use **fault domains** and **update domains** to protect workloads against host update failure and single points of hardware failure



Availability zones

Allow workloads to be deployed to different data centers in the same Azure region, protecting your workloads against data center failure

The latency between availability zones is low, and generally allows for **synchronous** data replication



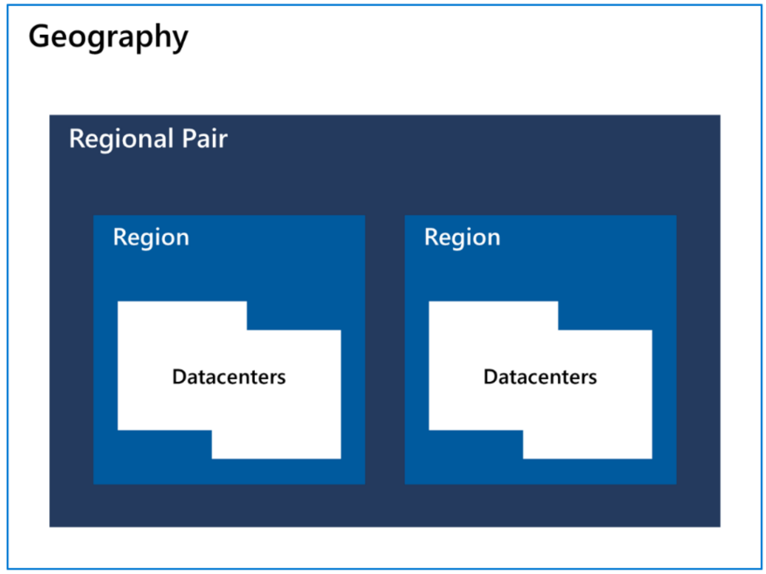
Paired Azure regions

All Azure regions have a "paired" region in which deployments and updates are applied

These paired regions are in the same geography

If a multi-region disaster happens, one region in each pair will be prioritized for recovery

Regional pairs are set and cannot be changed



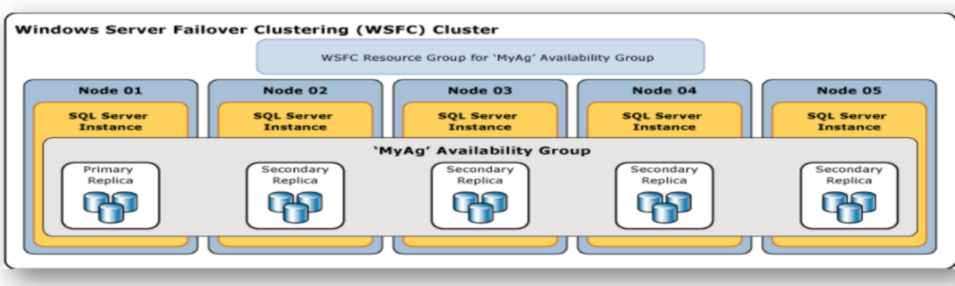
Always On Availability Groups

Provides HADR protection

Database transactions are committed to the primary replica, and then the transactions are sent either synchronously or asynchronously to all secondary replicas

If the secondary replicas are geographically separate - **asynchronous** availability mode is recommended.

If the replicas are within the same Azure region - **synchronous** commit can be used



Hyperscale SQL Database

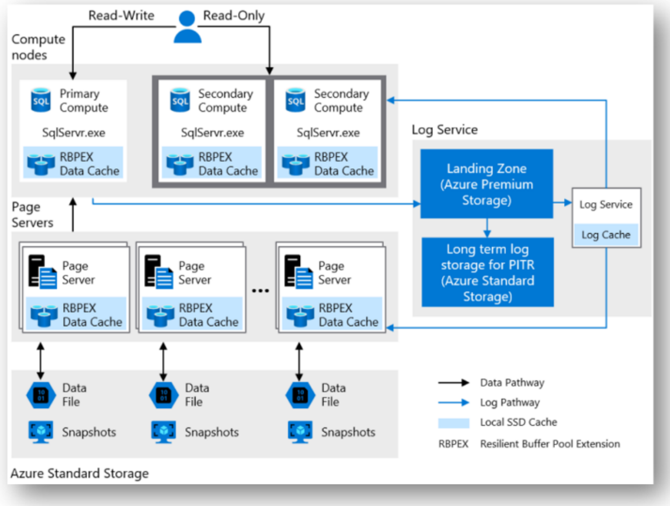
Designed for very large OLTP databases – as large as 100 TB - billed only for the capacity you use

Able to auto-scale independently storage and compute with no limits

Restores in minutes rather than hours and days

Scale up or down in real time to accommodate workload changes

Other databases are limited by the resources available in a single node, databases in the Hyperscale service tier have no such limits



Active geo-replication

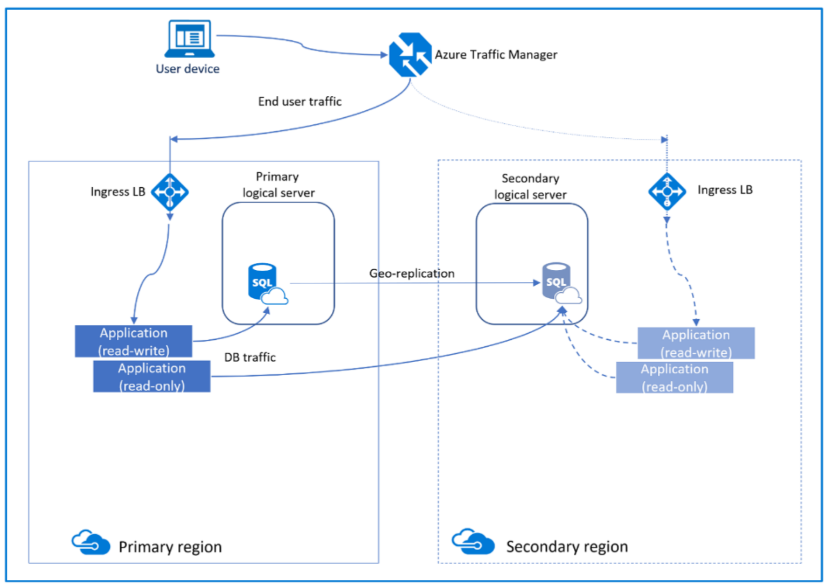
Provides automatic **asynchronous** replication of committed transactions on primary database to secondary database in the Azure region(s) of your choice using **snapshot isolation**.

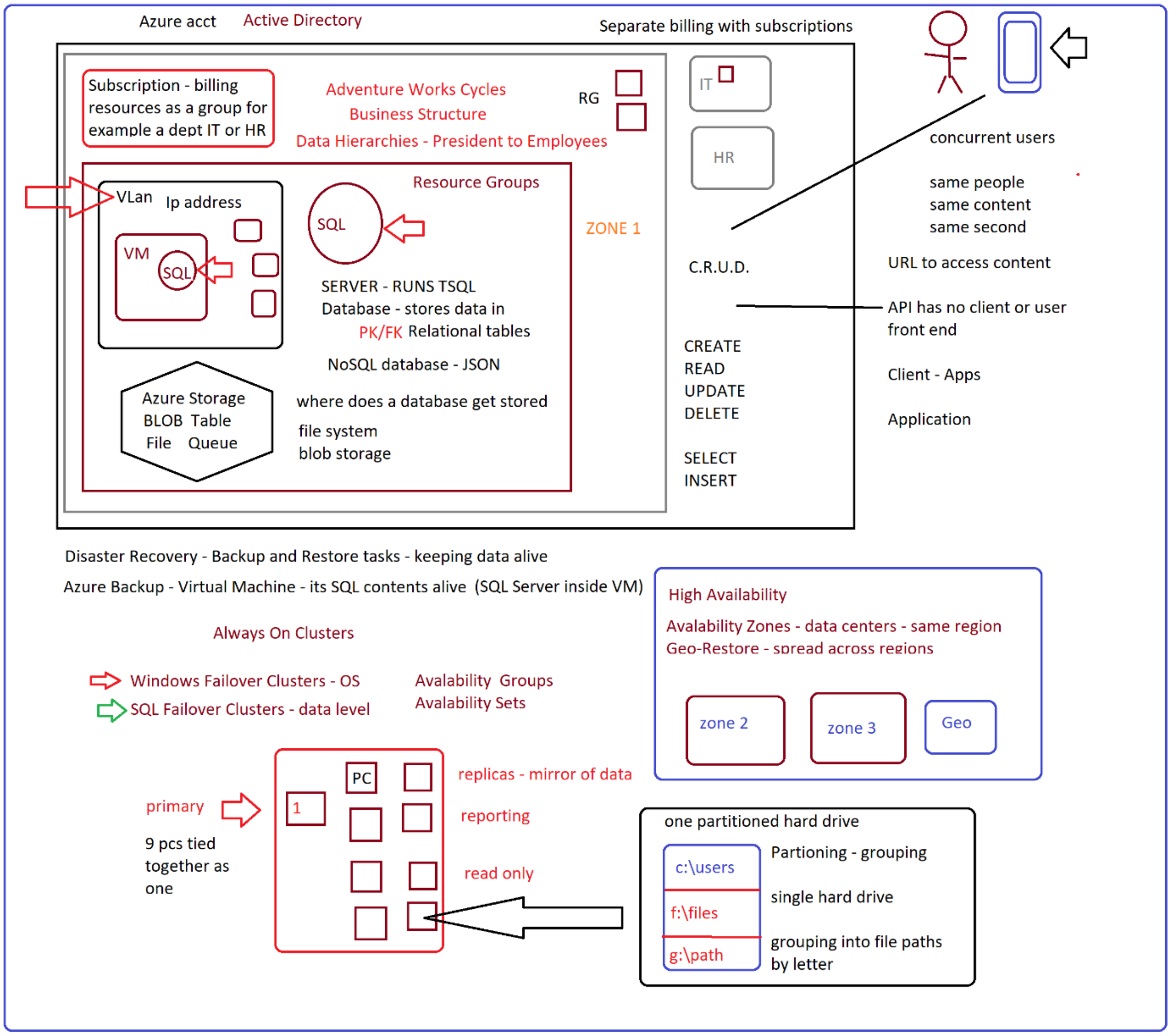
Utilizes **Always on Availability Groups** and **Active Geo-replication**

Provides a **readable secondary copy** of your database, no matter your service tier

You can have up to 4 secondary copies

Failover requires connection string updates





1. Relational Database
2. Microsoft SQL Server 2019/2022 Developer Free - Enterprise level
3. MySQL Linux Open Source SQL Server
4. Apple/Mac - Linux
5. 2 styles of development operating systems
6. Hybrid of MSoft & Open Source
7. Connecting different computers to each other -
   - Security - VNets or VLans - Range of IP Addresses - electronic box

8. Boundary - what or who can cross it?

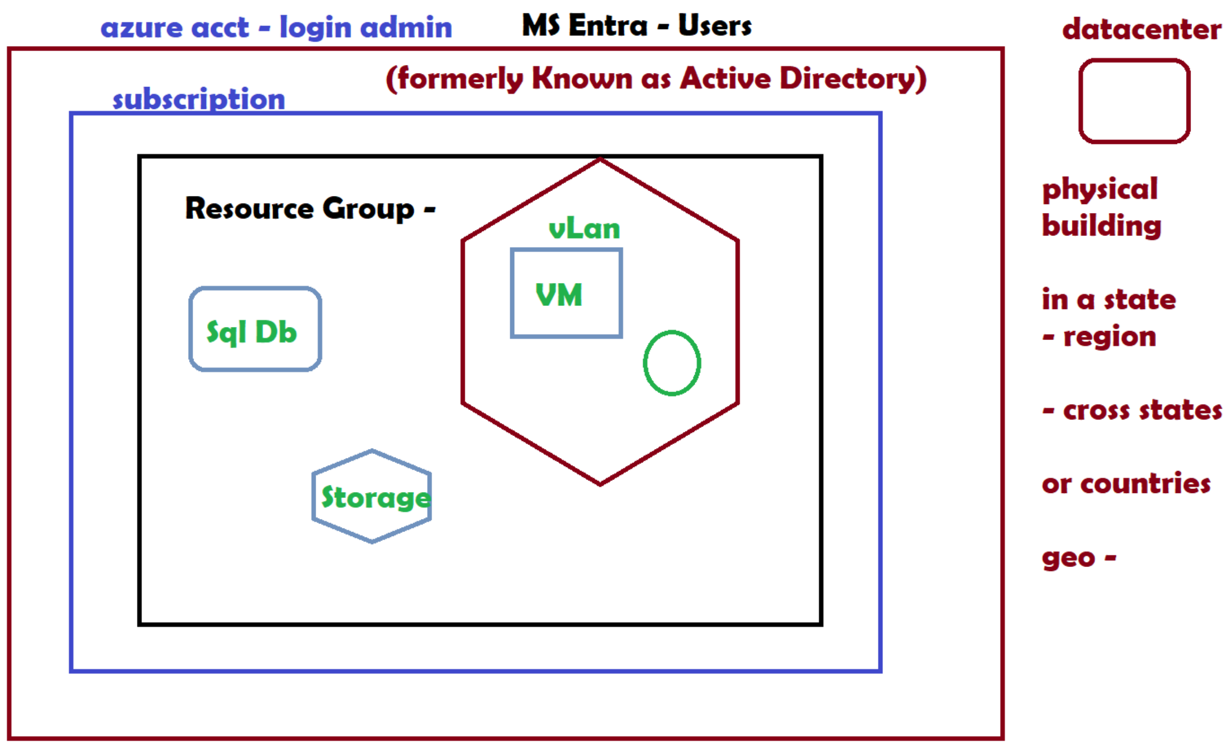
9. Domain
10. VLan

- I have a place at Azure
    - to store items
11. ---- Azure Blob Storage: files (Word, Excel, Picture, Movies) - maybe table (SQL Relational tables) data - maybe its JSON (NoSQL) data
    - Azure SQL Database - serverless -
    - VM- inside a Vlan with an installed SQL Server & SSMS
12. Message - Payload - the data being sent in a chat -
13. thc client side chat app assemble
14. text or not text - put in JSON or SQL format -
15. pictures - Blob (not Columns - Rows)
16. movie or music - media - Blob (not Columns - Rows)
17. Click Send
    - over the web to where? - (use IP addresses to find each device)
    - Security - HTTPS (SSL) - ?
18. __IOT - Internet of Things - Devices connecting to each other - wireless
19. ---- cell, tablets, appliances, tvs, clocks, Google Nest, Security Like Baby cams -
    - Ring - mothing detectors - doorbell cameras - EVehicles -
20. text or not text - put in JSON - NoSQL or SQL format -
21. Azure Hosting Service - Resources can be turned into JSON Azure Resource Manager Template
22. Riverbed - message that flows through your systems
    - To Store for
    - short or
    - longer times - History - Analyzing - SQL Relational Servers Systems - Maybe Table Storage
23. --- Features to help manage History or how long something lives in storage -
24. --- Products/Resources to use to store or queue or react with code to manage - alert - ?
25. __ DevOps and Automate everything I can with Azure Resource Manager Template & GitHub
26. __Hot - often accessed - Data Tier - for billing
27. __Cold - not often accessed
    - The steps needed to move your data - files from the source
    - On Premise - a physical builing your business owns
    - Off Premise - Migrate to Azure -
        - Steps or jobs or actions - tasks to do
28. --- What tool to use -
29. --- Am I allowed to connect and do the work
30. --- What are the Governance or business rules I must follow
    - Health Insurance
    - Government
    - Data Classification - using AI to classify data - makes features in the software work.

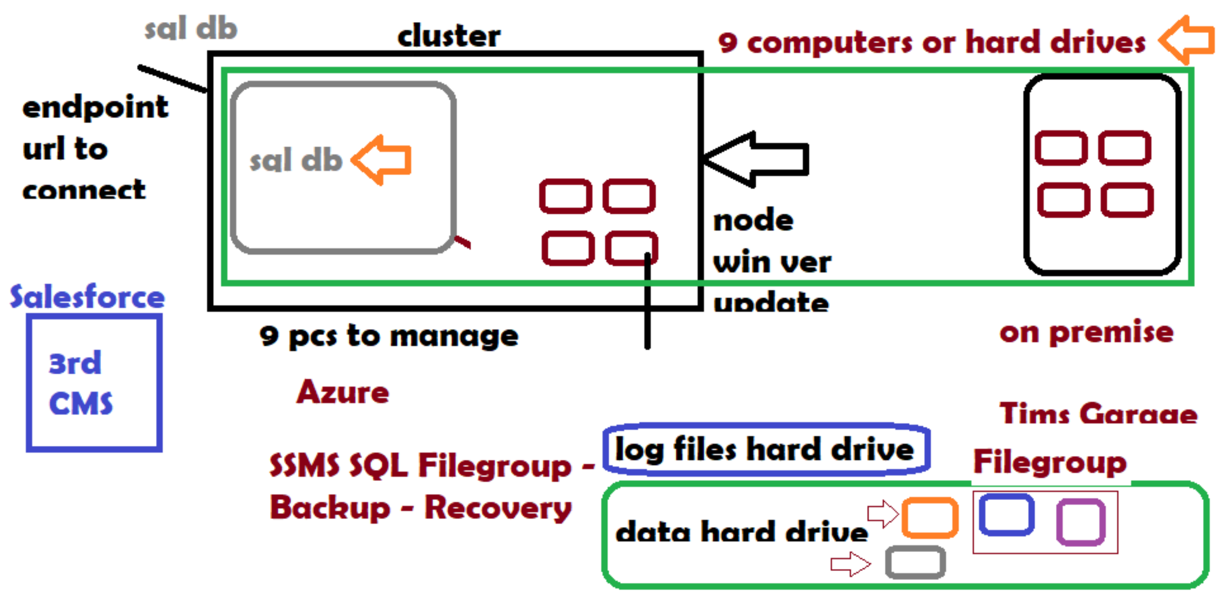
    - Maps - Geography data
    - Numbers w/ ledgers - Accounting data

    - contain Personally Identifiable Information (PII) data (RLS - Row Level Secretary)

31. __Scaling
    - growing resources to manage more or less
    - storage
    - number of concurrent users
        - Frequency - hot or cold (History - Analyzing)
    - - Modified Date Column
32. Send a print job to your printer -
    - send it to a queue -
        - hold a file - first in - first out
    - until the printer is ready
33. one printer - many print jobs
34. id number generated by the machines - keys
35. used to secure one machine to another
36. encrypt or create a math algorith to hide data (data encryption key)
37. Least privilege - RBAC - user is put in a group that has the permissions to whatever (EXECUTE AS)
38. Layer between user and data - so you cannot get hacked
39. Active Directory - MS Entra ID
40. Store encryption keys in Azure Key Vault.
41. Retain backups of the PII data for two months - Soft Delete
42. Encrypt the PII data
43. at rest - encryption
44. in transit - encryption
    - Secure Socket Layer SSL - http(s) - the connection between machines but content
45. and in use - encryption
46. Disaster Recovery - did I lose data - how much over time in minutes Point In Time Recovery
47. 99.99% SLA - I will never crash or lose data
        - some where else in between -
48. Rebuilding your business if it fails
    - power goes out
    - hard drive crashes - backup
    - natural disaster - something large enough to knock down a building - hurricane -
49. database has at least 5 physical files
50. 30,000 database * 5 = 150 files being accessed by 160000 users every second
51. every database must live on a separate SQL Server - (1 server can have many databases)
52. 30 servers - manage
53. column names in this database are different from the target database
54. spreadsheet - FirstName - fName
55. database replica - mirror database https://learn.microsoft.com/en-us/training/modules/deploy-paas-solutions-with-azure-sql/2-explain-paas-options-deploy-sql-server-azure



56.



57.
58. Loose Text Notes
59. Azure Bastion is a fully managed service that provides more secure and seamless Remote Desktop Protocol (RDP) and Secure Shell Protocol (SSH) access to virtual machines (VMs) without exposure through public IP addresses.
60. https://azure.microsoft.com/en-us/products/azure-bastion
61.

62. IP Addresses - Firewall Rules - Internet Protocol - Hypertext Transfrer Protocol or HTTP - connect things together
63.
64.
65. Open Source means no one owns the code or the product
66.
67. MySQL
68. Linux - OS
69. -- Branded as IoS
70.
71. Free to download and use
72.
73.
74. ip address - global network
75.
76. 192.168.10.10 - address of my pc for example
77.
78. Group them in a Virtual Network
79. - Subnet a group in a VLAN
80. -- VLAN can have many Subnets
81.
82. Behaves like a fence and conrolls user access with Firewall Rules
83.
84. Service Level Agreements - SLA
85.
86. Consultant Design an Azure Solution App
87.
88. Bank legal agreement MS to keep data alive 99.9
89.
90.
91.
92. Analytics on data
93.
94. Transform your data
95.
96. combine any data
97.
98. build and deploy models at scale
99.
100. PBI Desktop Report
101.
102. ETL -
103. Extract Transform Load
104.

what can a machine store as data that we can use

columns and rows - tables

raw data
- textfile.txt
- JSON
- NoSQL Servers

structure
MyStructuredData.csv
- comma separated values

Column headers

row data

1 + 1 = 2
- data type number

if the 2 values are numeric the plus operator will do math

alphabet or words or chinese character set
- pc knows nothing
- - convert to numbers binary code - 11110000

tim + 1 = tim1
- plus operator concatenates not do math
- writes a sentence
- data type is string

continuous integration and continuous deployment (CI/CD)

pipeline

MSoft
SQL Server 2012
- ran only on MSoft Windows Server
-- both had an SLA

In 2016 MSoft Open Sourced Dotnet Framework
- SQL server now runs on Linux

Virtual Machine
- a remote computer running an OS
-- Hypervisor software runs VM's

Docker containers
- A way of deploying anything to another computer and have it run what is inside it.

-- No Operating system
--- Install Docker to run containers

Connect to everything with URLS like web pages do.

image
-- when is it not a .jpg file or .bmp a picture from a phone

when we package files or whatever to often use as a template
-- re use the template
-- saves time

files with .iso or .img extension

Project Management Steps

Daily Scrum Meeting of the Team

API Server side
- expected you build the Client app

Google Maps
- Builds the map based information you give it.

-- You build the phone app to show it.

MS Entra ID
- after Open Source
-- Uses OAUTH

-- Open Authorization

Active Directory
-- MSoft User Directory
--- not built on OAUTH

I Owned and Licensed
2012 SQL & Windows
Users are in AD

move to Azure

What are these products

Store User Profiles in a database

- Allow to Create Groups or Roles
-- Assign Permissions to Groups or Roles

RBAC
Role Based Access Control

Go Daddy
- Domain Names -
- Server Certificate

First Azure account Email Address is ?

Office365
- Create Directory
-- User Tools - Office etc
--- SLA Per user Seat

Azure Data or App Side Azure
Parent MS Entra ID Directory
-- Add Users Delegate

bill together all resources
- default subscription

Web Site
Domain Name

User managements & access

Adventureworks
- Sell Bicycles

what kind of data or tasks happen

Retail Sales

Office365
- Tools for Users

Azure Hosting
- Servers Side Things
-- Data or Apps or API's

Front Facing Web Site
- Go Daddy Hosting Services
-- Wordpress & MySQL

MyBusinessName.com

Compute Resource - VM - Tablet

PC's
Hardware

Hard drive - Storage
RAM - Memory
OS - Operating System

Does Azure manage it completely or not?

Serverless platform does maintenance

-- More Out of the Box or Template
- Create and Connect and use it.

proper VM
sizing, hardware

a resource in a RG

storage service
networking service

Create Adventureworks In Cloud
domain name .com

365 - Azure

Website lives somewhere else than Azure for now - 100.00 year

Budget 25000.00 a year

What do I need?
What do I have already
- 5 pc's - win 10
- 1 SQL Server 2019 on one of my 5 pc's

Migrate to Azure

What moves
- Data
How do I want to do it?

Can I copy the entire database?
- Backup - Restore

Move it to Azure Storage for Archiving but not online use.

VM w SQL installed
- Backup to another SQL Server

Basic Setup?

Create a Web App Service
- Web Server that runs code
Host my Wordpress Web Site

-- Connect to Azure Resource in my MS Entra ID Directory

Infrastructure - managed more parts

325. Platform - Serverless
326. - Azure SQL Database
327. - Connection API's
328. --- assemble different kinds of data systems
329.
330.
331.
332. availability/redundancy, is it possible to run a database VM in one region and mirror it on a different region
333.
334. deploy SQL Server in Azure
335.
336. Provision a Service
337.
338. SQL Server
339.
340. - database file itself - has a version
341. 2012 - use 2012 SQL DB Engine to access it
342.
343. the actual file is not Queryable
344. MSoft 2012 - 2022
345. Oracle
346. MySQL
347.
348.
349. LMS
350. CMS
351. Learning Management System
352. Content Management System
353.
354. Wordpress
355. Client - web based - html - Javascript
356. - MySQL for database
357.
358. Salesforce
359.
360. Human Capital Management HCM
361.
362.
363.
364. compute optimized
365.
366. computer with a good CPU and RAM config
367.

368. CPU - Central Processing Unit - chip
369.
370. do math
371.
372. open the hard drive
373.
374. write to RAM
375.
376. do the work
377.
378.
379. Video - very CPU expensive - intensive
380.
381. GPU
382.
383. Graphics Processing Units

Class Notes

Azure - where the database & SQL server may be hosted or lives

Database - the file or storage of columns and rows to access and do C.R.U.D. Operations within relational structured tables

Administrator - the job you might do

- allow access to databases and other things

RBAC

- Role Based Access Control

Data Fundamentals

- understanding table and rows in an SQL table

-- Normalize is the word to describe creating the relationships and tables

Anything you type in any language

- instructions to a machine

-- execute line by line

---- Flow to everything - DevOps - Developer Operation ------ Automate everything - CI/CD - Continuous Integration and Continuous Deployment

Tools to do the job

- Power Automate
- Power Point
- Power BI

API - Application Programming Interface

SERVER - for us Azure or Your Garage Business

-- How do I connect to something

-- URL address

--- Name of a computer or software SQL Server name

---- Credential or GUID - called Registration of your app

Client Side - Chat App - Client Side

- What does every app have

--- SQL Server - Tables in relational storage

--- Represent Context of: A Person, A Plane, A Bike

--- An Object in Computer Languages

----- JSON: JavaScript Object Notation

{

name:value pair

key:value pair

fName:tim

column:row

}

Querystring in a URL

https://myDomainName.com/home.page/?guid=123456789

C.R.U.D. Operations

- Read and write to hard drive somewhere

- Storage - C: drive etc - some place to put things

-- If a message - traffic - the querystring - the JSON is not stored

-- It is handled like a printer Queue

---- In order recieved then sent out - (uber eats)

What can a human do with a machine

- data movement from client to server and back

--- Round trip - Request - Response over HTTP to get content or data

execute means Run or EXECUTE AS could be F5 keyboard - code - script - any language the instructions

Azure Functions - app that runs code at Azure - for whatever work you need to do at the platform

C - CREATE - INSERT

R - READ - SELECT

U - UPDATE - UPDATE

D - DELETE - DELETE

- Today we Datawarehouse not delete

To find any Trend or Analyze and data

-- History - duplicate data

OLAP - Online Analytical Processing - DW

OLTP - Online Transactional Processing - Producion database

at Azure - each operation could be billable

Hosting as a Service - HaaS

A place to put things

-- Electronic Real Estate -

data

- Structured - SQL

- unstructured - JSON - NoSQL

- files - docs, .pbix, mytext.txt, myDatabse.mdb, myDatabase_Log.ldf

Once I have an Environment

- Azure Resources

- Virtual Machine - Window OS

- SQL Server installed

-- Connect with RDP - Remote Desktop Protocol

Connects to the another computer - a Virtual Machine

Then

User Tools

- Connect to SQL Server for tasks

-- SSMS: SQL Sever Management Studio

--- Data Explorer

Tasks or user workflow

- like climbing a ladder

series of steps

Project Management: Agile

is it Microsoft SQL or MySQL

or NoSQL

It DOES NOT MATTER

record that represents

tims order

from dominos

for a large Pep Pizza

Deliver

POS billing

Microsoft - Owns

- Windows 10, 11 SQL Server

-- Versions and editions

--- Requires License - SLA

Open Source or 3rd Party

-- No One Owns the Code

OS - Linux

SQL - MySQL - Executes SQL language in relational tables

Apple - rebrands linux as IoS

Amazon - Sells things

Facebook - Wordpress on Steriods

- Chat app

- Group people as Friends

-- Like things - generate analytic metric numbers

---- text message

----- upload video or pictures

Free

MyAPI does this:

write code and store data

SQL

Wordpress

- Client - Server app

-- Uses HTML for client side - web pages

--- MySQL for server side

What does the user run my app with?

Android

IoS

Web Browser

- ?

something I connect my app to

Google Maps API

Azure SQL Database

Web page that shows a google map

connect to google with a url in my web page

Google tracks me with GPS

- Longitude - 1.0
- Latitude - 2.0

Location on a map

App

Prog

Connector

API may be called a Service

Application

- SQL server store data

- Programming
- CRUD code to read and write data
- show to User in an app

- Interface - Client App

MyChat app for Android

- code it in C#

--- Connect to the Chat API

Web Browser runs in all OS

- Runs everywhere

IoS Apple Store

- Sunn Systems looks like C# which looks like JavaScript - using curly braces

API JOB Code that Manages the Chat

predetermined list of

DevOps

tasks and workflows that are assembled together to

Part of an API

simplify communication

- URL

Server Side - Azure or AWS - the cloud we connnect to

- application always has a database

Iot - Internet of Things

Cell Phone

Tablet

PC

devices that connects

Drones

Wi-Fi - Smart things

Refrigerator

Web Server takes the url request from the client browser for MS Learn

- Take the learn.MS.com turn it into an ip address connected to a file location like c:\myFolder

198.168.10.101 point to c:\myFolder

https://learn.microsoft.com/en-us/credentials/certifications/azure-database-administrator-associate/

?

practice-assessment-type

=

certification

My API Does this

- Finds files in my servers
- open & close files
- C.R.U.D. Ops

Execute application code to do something

- write a record to a row

Send response back to the Client

Chat App

Define the steps or tasks

user - click or drag whatever to make it work

app or the api steps

App

User

- Sign in
- To address Send
- From address
- chat text

- pic or video

know how to connect to the api

Server Side or the API

- what happens to the user request -

-- access SQL data - Storage - a Web Page ???

Assemble the Response

what kind of information or data is my app using

SQL

- whose MSoft or not?

columns and rows

- millions of records that

take up hard drive space

represents money to a business

On and Off Premise

Off Premise means you own it without paying for access to it

Blob

Binary Large Object

OOB

Out of the box

ARM Template

Azure Resource Manager file

- JSON format saved as text.

Bicep to deploy ARM Templates

SQL Server Software - Free Developer Edition

Jobs it can do

SQL Database Engine

- Work with tables with T-SQL

Other Features I could turn on

SQL Server Analysis Services (SSAS)

- before Power BI Data Tools
- use a data cube as storage

SQL Server Integration Services (SSIS)

- ETL JOBs
- Moving data on a schedule

SQL Server Reporting Services (SSRS)

images means a version of software

- Windows Server 2022 and SQL Server 2019

Disaster Recovery

- Schedule copies of data and log files to be made

Backup Copy

- Scheduled over time
- have many parts or files

Restoring a database

- reassemble the backup files in correct order

https://learn.microsoft.com/en-us/training/modules/prepare-to-maintain-sql-databases-azure/3-understand-sql-server-azure-virtual-machine

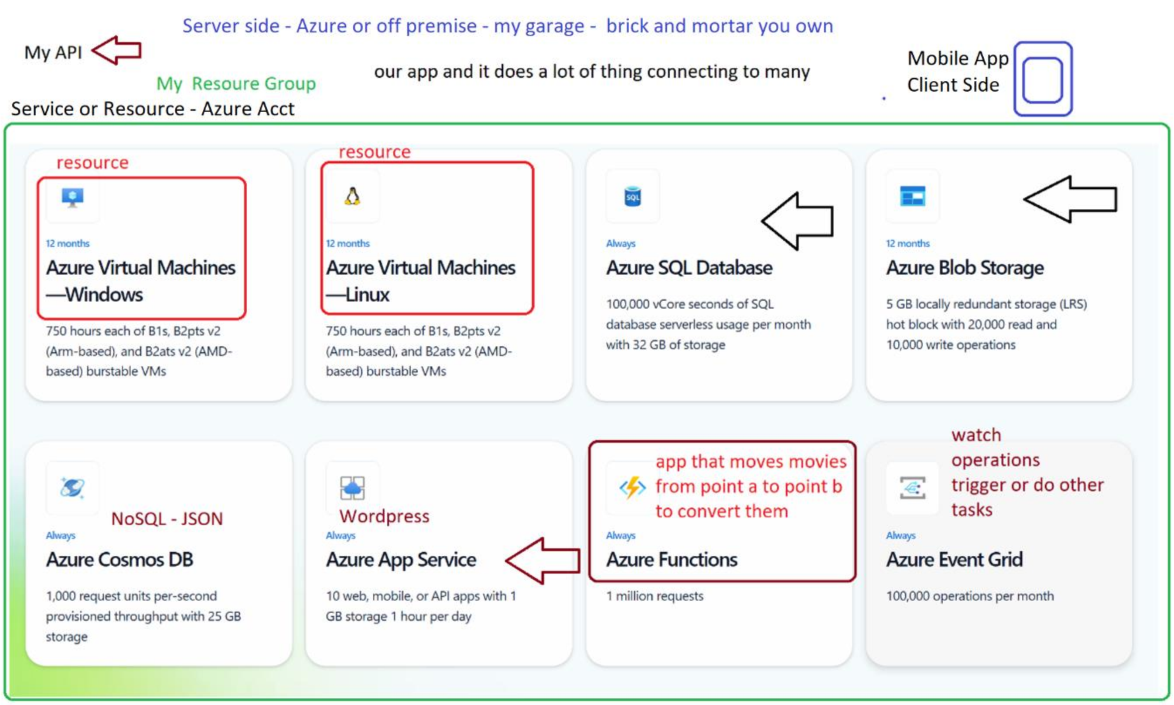
features to support high availability including availability sets,

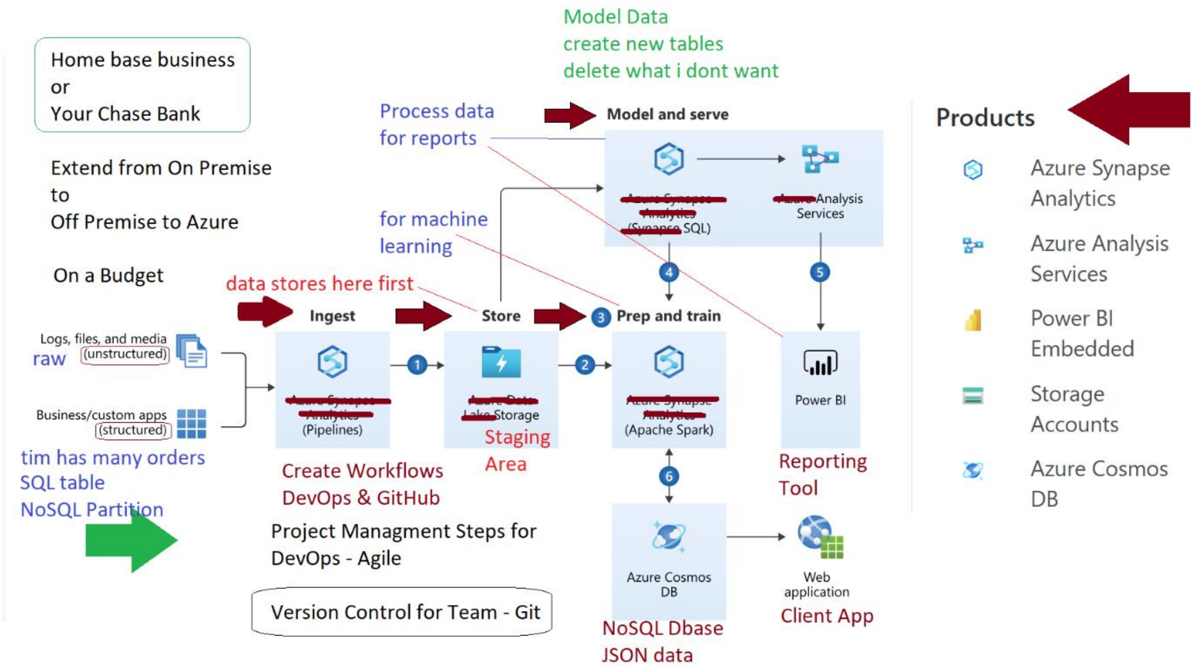
availability zones,

and load-balancing techniques that provide high

- distributing incoming traffic

among Virtual Machines.





business - Adventureworks is Retail Bicycle Sales

tabs at bottom

what tables do I need
or have
what PK & FK are there?

Dept

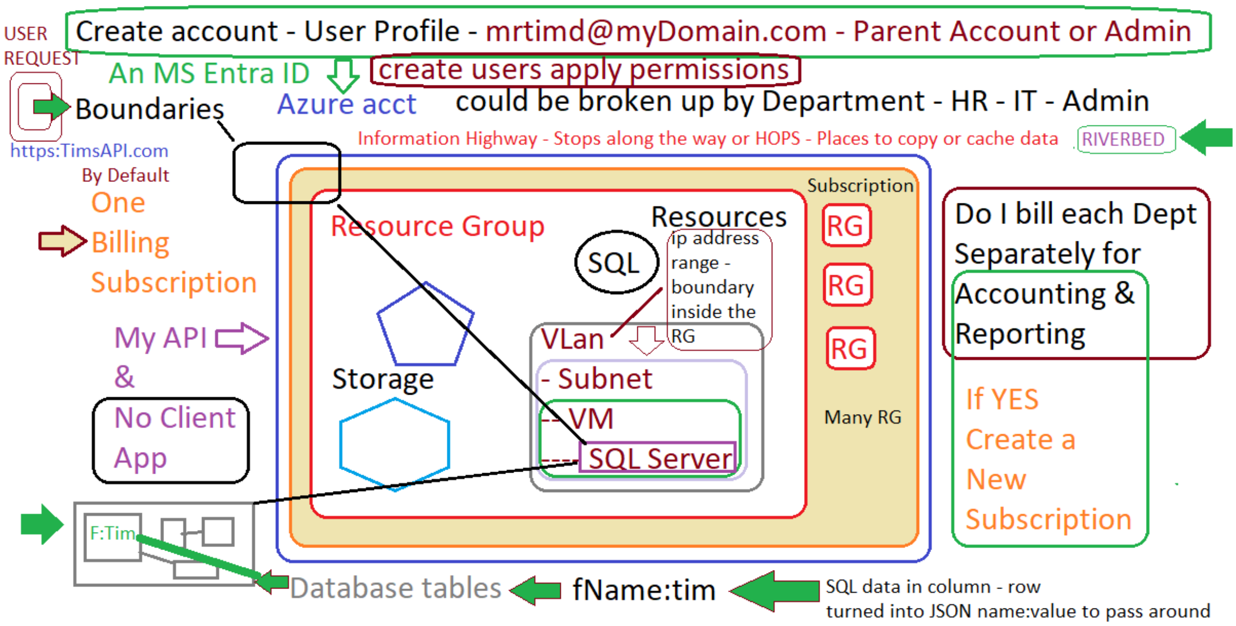
logical groups of
information

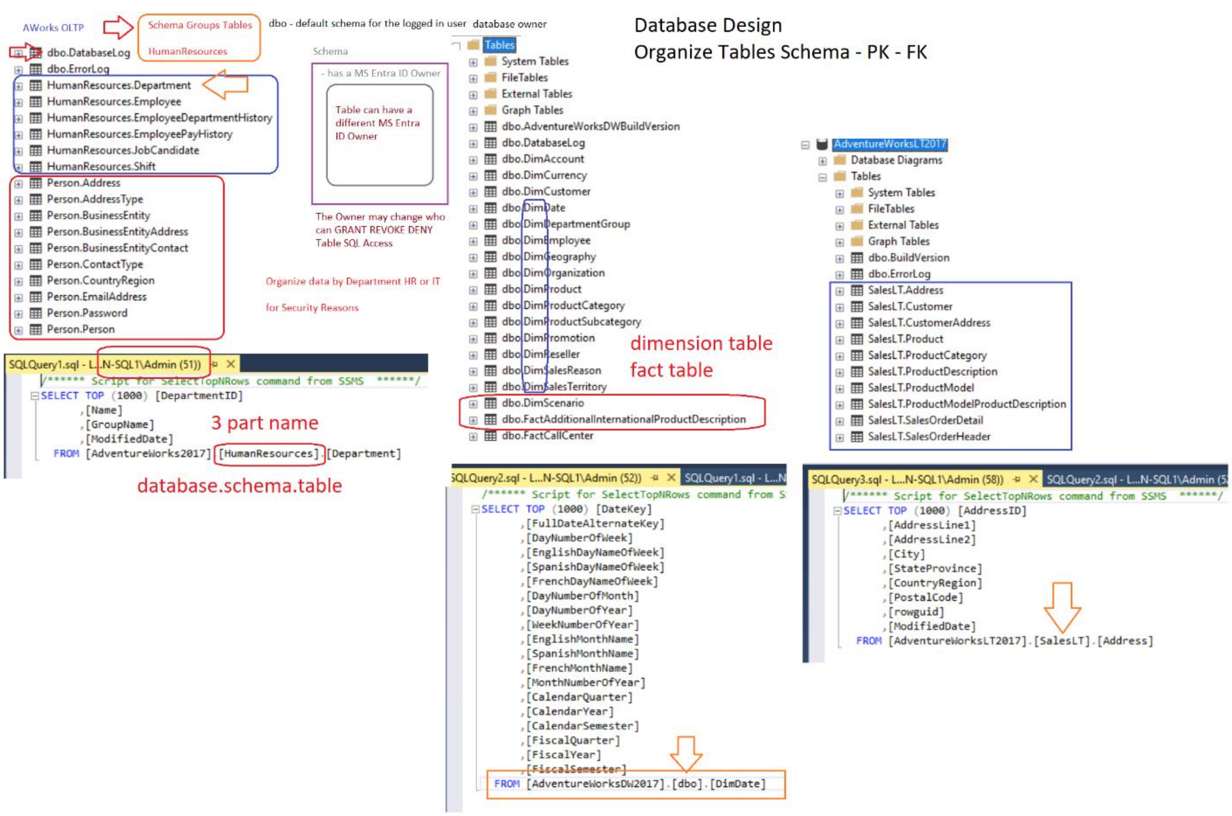
HR
Admin
IT

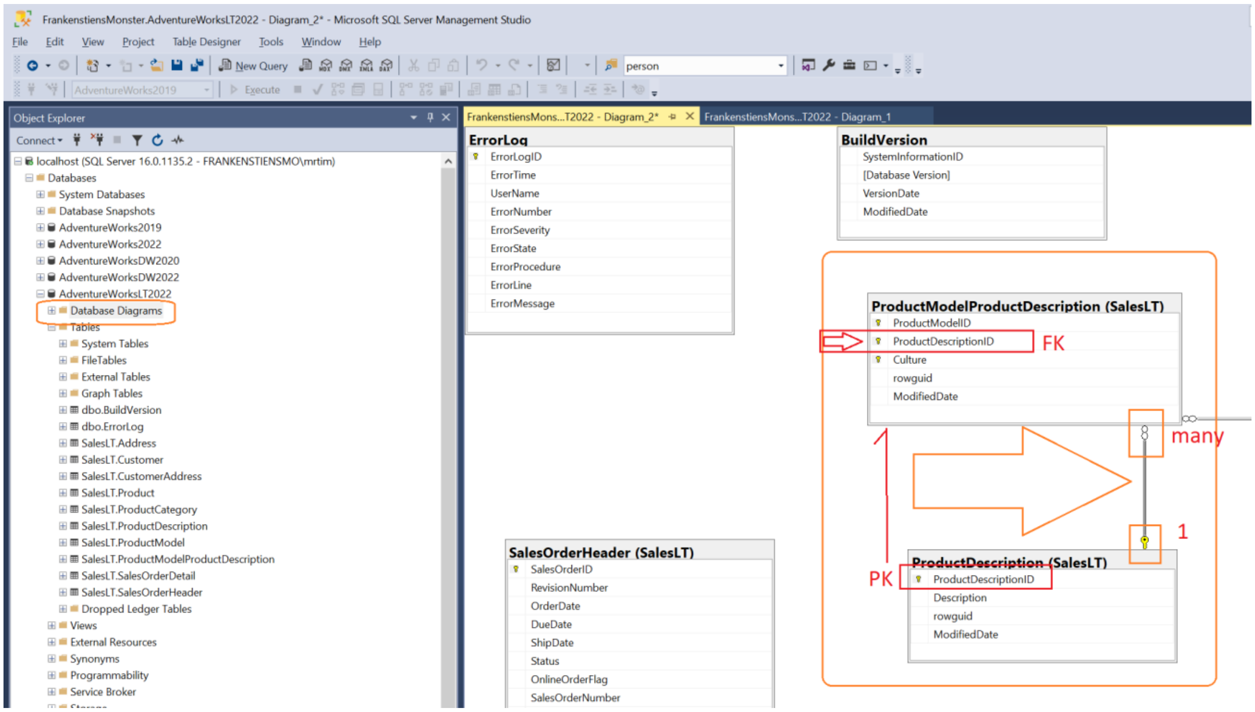
sales
customers
money
products

employees

CEO
VP
Dept Head
Programmers
Part Time


USER REQUEST
Create account - User Profile - mrtimd@myDomain.com - Parent Account or Admin
An MS Entra ID
create users apply permissions
Boundaries
Azure acct
could be broken up by Department - HR - IT - Admin
https:TimsAPI.com
Information Highway - Stops along the way or HOPS - Places to copy or cache data
RIVERBED
By Default
One
Billing
Subscription
My API
&
No Client App
Subscription
Resource Group
Resources
SQL
ip address range - boundary inside the RG
Storage
VLan
- Subnet
-- VM
--- SQL Server
RG
RG
RG
Many RG
Do I bill each Dept Separately for Accounting & Reporting
If YES Create a New Subscription
F:Tim
Database tables
fName:tim
SQL data in column - row turned into JSON name:value to pass around

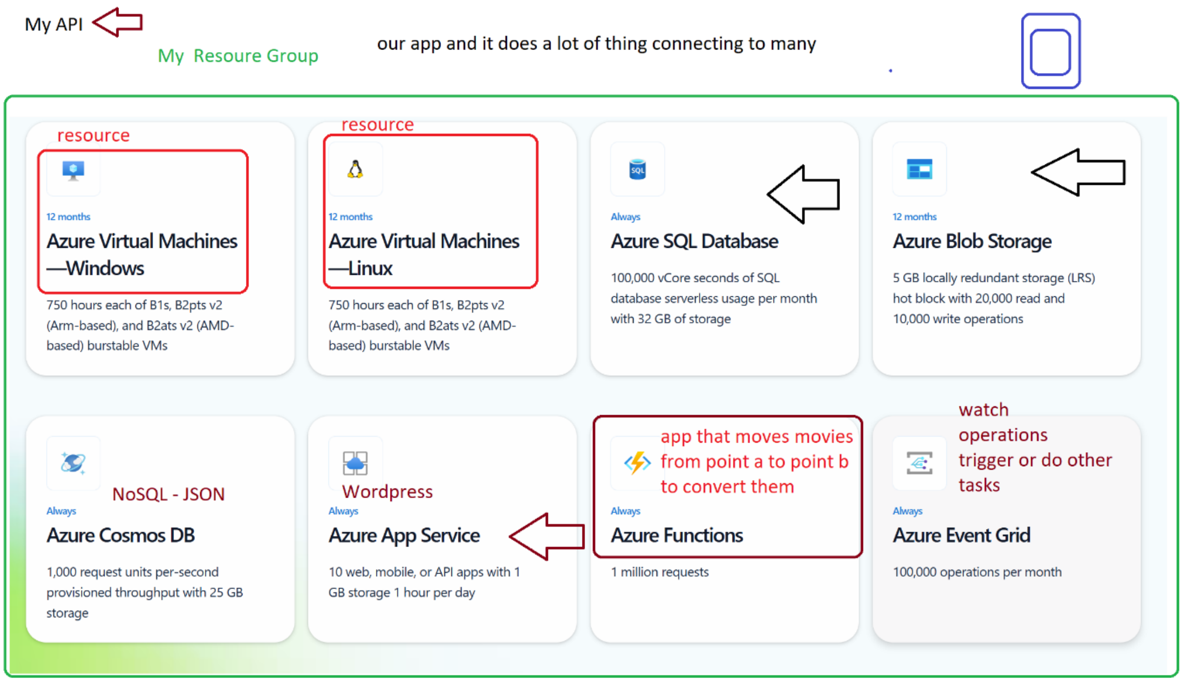

AWorks OLTP
Schema Groups Tables
HumanResources
dbo - default schema for the logged in user
database owner
Schema
- has a MS Entra ID Owner
Table can have a different MS Entra ID Owner
The Owner may change who can GRANT REVOKE DENY Table SQL Access
Organize data by Department HR or IT
for Security Reasons
Database Design
Organize Tables Schema - PK - FK
dbo.DatabaseLog
dbo.ErrorLog
HumanResources.Department
HumanResources.Employee
HumanResources.EmployeeDepartmentHistory
HumanResources.EmployeePayHistory
HumanResources.JobCandidate
HumanResources.Shift
Person.Address
Person.AddressType
Person.BusinessEntity
Person.BusinessEntityAddress
Person.BusinessEntityContact
Person.ContactType
Person.CountryRegion
Person.EmailAddress
Person.Password
Person.Person
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.AdventureWorksDWBuildVersion
dbo.DatabaseLog
dbo.DimAccount
dbo.DimCurrency
dbo.DimCustomer
dbo.DimDate
dbo.DimDepartmentGroup
dbo.DimEmployee
dbo.DimGeography
dbo.DimOrganization
dbo.DimProduct
dbo.DimProductCategory
dbo.DimProductSubcategory
dbo.DimPromotion
dbo.DimReseller
dbo.DimSalesReason
dbo.DimSalesTerritory
dbo.DimScenario
dbo.FactAdditionalInternationalProductDescription
dbo.FactCallCenter
dimension table
fact table
AdventureWorksLT2017
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.BuildVersion
dbo.ErrorLog
SalesLT.Address
SalesLT.Customer
SalesLT.CustomerAddress
SalesLT.Product
SalesLT.ProductCategory
SalesLT.ProductDescription
SalesLT.ProductModel
SalesLT.ProductModelProductDescription
SalesLT.SalesOrderDetail
SalesLT.SalesOrderHeader
SQLQuery1.sql - L...N-SQL1\Admin (51))
/****** Script for SelectTopNRows command from SSMS ******/
SELECT TOP (1000) [DepartmentID]
,[Name]
,[GroupName]
,[ModifiedDate]
FROM [AdventureWorks2017].[HumanResources].[Department]
3 part name
database.schema.table
SQLQuery2.sql - L...N-SQL1\Admin (52))
/****** Script for SelectTopNRows command from S
SELECT TOP (1000) [DateKey]
,[FullDateAlternateKey]
,[DayNumberOfWeek]
,[EnglishDayNameOfWeek]
,[SpanishDayNameOfWeek]
,[FrenchDayNameOfWeek]
,[DayNumberOfMonth]
,[DayNumberOfYear]
,[WeekNumberOfYear]
,[EnglishMonthName]
,[SpanishMonthName]
,[FrenchMonthName]
,[MonthNumberOfYear]
,[CalendarQuarter]
,[CalendarYear]
,[CalendarSemester]
,[FiscalQuarter]
,[FiscalYear]
,[FiscalSemester]
FROM [AdventureWorksDW2017].[dbo].[DimDate]
SQLQuery3.sql - L...N-SQL1\Admin (58))
/****** Script for SelectTopNRows command from SSMS ******/
SELECT TOP (1000) [AddressID]
,[AddressLine1]
,[AddressLine2]
,[City]
,[StateProvince]
,[CountryRegion]
,[PostalCode]
,[rowguid]
,[ModifiedDate]
FROM [AdventureWorksLT2017].[SalesLT].[Address]


FrankenstiensMonster.AdventureWorksLT2022 - Diagram_2* - Microsoft SQL Server Management Studio
File Edit View Project Table Designer Tools Window Help
New Query
person
AdventureWorks2019
Execute
Object Explorer
Connect
localhost (SQL Server 16.0.1135.2 - FRANKENSTIENSMO\mrtim)
Databases
System Databases
Database Snapshots
AdventureWorks2019
AdventureWorks2022
AdventureWorksDW2020
AdventureWorksDW2022
AdventureWorksLT2022
Database Diagrams
Tables
System Tables
FileTables
External Tables
Graph Tables
dbo.BuildVersion
dbo.ErrorLog
SalesLT.Address
SalesLT.Customer
SalesLT.CustomerAddress
SalesLT.Product
SalesLT.ProductCategory
SalesLT.ProductDescription
SalesLT.ProductModel
SalesLT.ProductModelProductDescription
SalesLT.SalesOrderDetail
SalesLT.SalesOrderHeader
Dropped Ledger Tables
Views
External Resources
Synonyms
Programmability
Service Broker
FrankenstiensMons...T2022 - Diagram_2*
FrankenstiensMons...T2022 - Diagram_1
ErrorLog
ErrorLogID
ErrorTime
UserName
ErrorNumber
ErrorSeverity
ErrorState
ErrorProcedure
ErrorLine
ErrorMessage
BuildVersion
SystemInformationID
[Database Version]
VersionDate
ModifiedDate
ProductModelProductDescription (SalesLT)
ProductModelID
ProductDescriptionID
Culture
rowguid
ModifiedDate
FK
many
1
PK
SalesOrderHeader (SalesLT)
SalesOrderID
RevisionNumber
OrderDate
DueDate
ShipDate
Status
OnlineOrderFlag
SalesOrderNumber
ProductDescription (SalesLT)
ProductDescriptionID
Description
rowguid
ModifiedDate

Reference

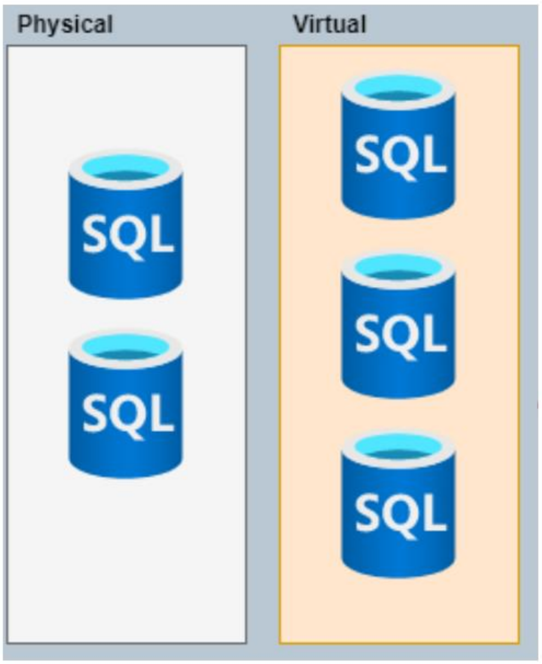
1. Applies To what product SQL
   a. [SQL Server docs navigation tips - SQL Server | Microsoft Learn]

[Pricing - Azure SQL Database Single Database | Microsoft Azure]



Developer Sandbox
Win 10 Pro
Hyper - v to create VMs Locally

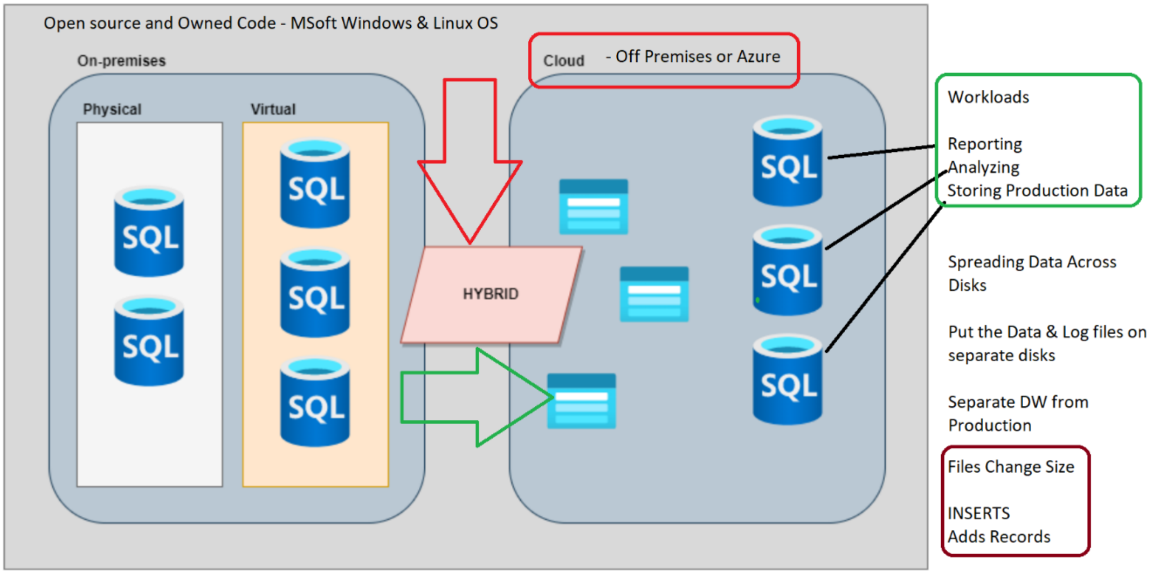
get free SQL dev version - install it 5 locations



My PC is the Host -

Everything shares CPU & RAM & Hard Drive

Table
Columns & Rows
One row has 20 columns - 5 columns store 2kb - 15 columns that store 20kb -
- Each row 350kb * how many rows = size table - project size based on Batch or number of records to INSERT



Table
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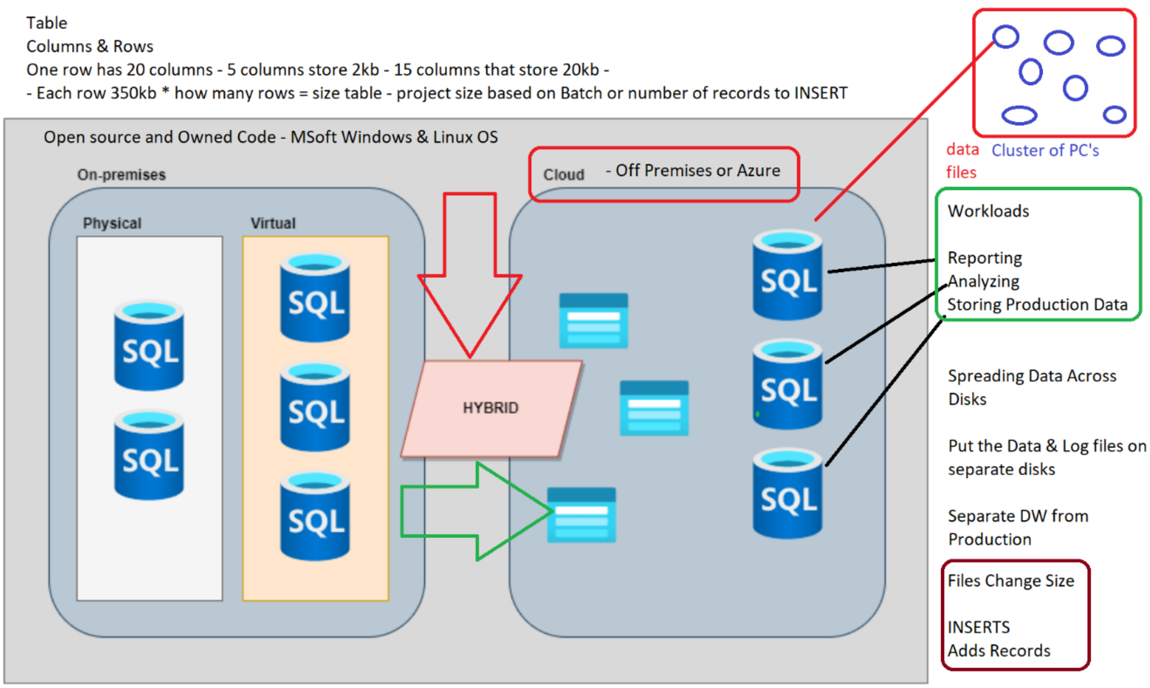


CI/CD - Every step of Each Process is Defined - Schedule Deployments

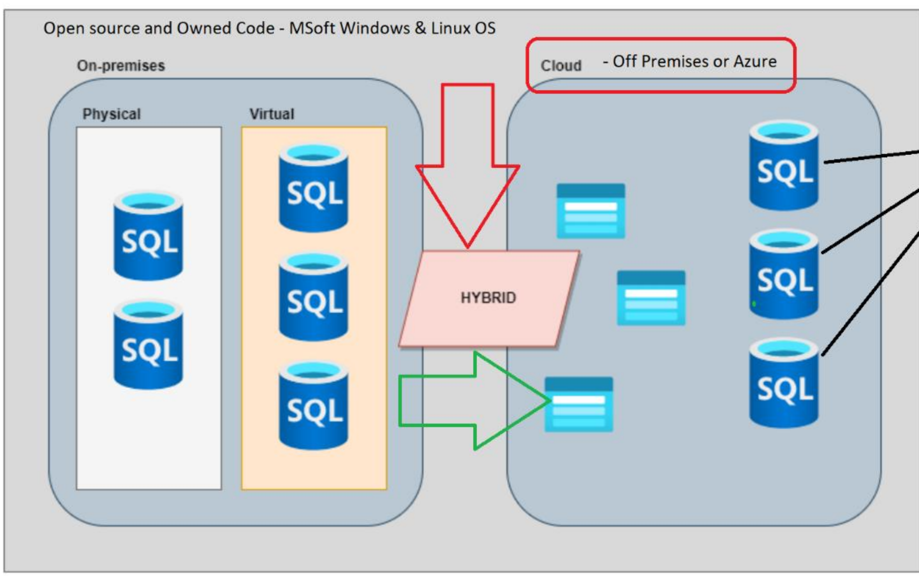
azure acct
Centrally Manages

external services
Github
Git Action
Move FIles
coding actions to move files
files live here
version 0.0
clone or fork - v0.1
Update

My App
V0.0
Feedback
v0.1 or 1.0

20 c# files that make it run
Assigned to a
Team Member
Work on the App
3 files to code or fix

Azure Pipelines
Repos
Repository a Versioned Place for files
DevOps
Boards
Allow the Boss to assign Work as Flow and Watch It - Start - In Prog - Done



Workloads

Reporting
Analyzing
Storing Production Data

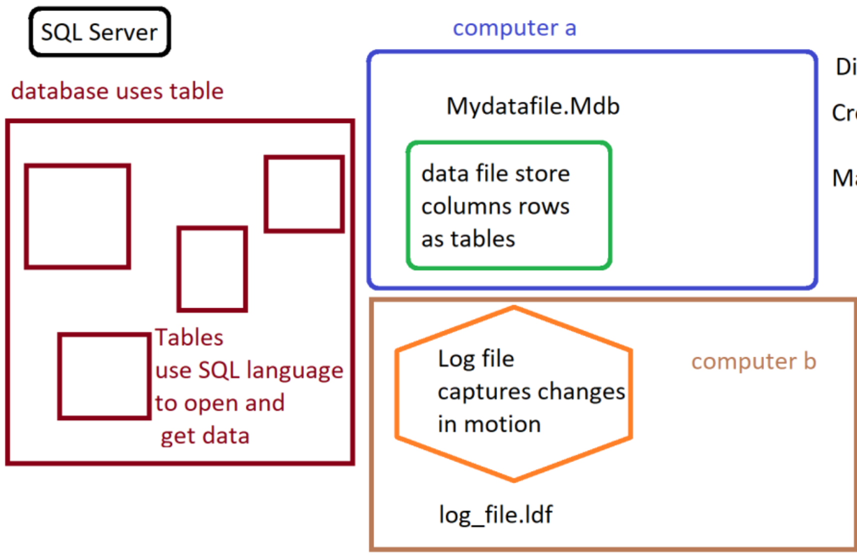
Spreading Data Across Disks

Put the Data & Log files on separate disks

Separate DW from Production

Files Change Size

INSERTS
Adds Records



Disaster Recover

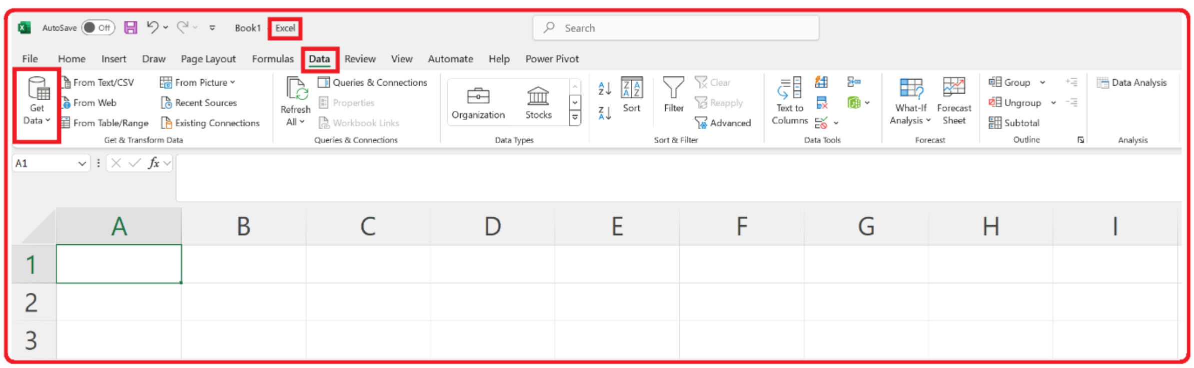
Create data file copies on schedule

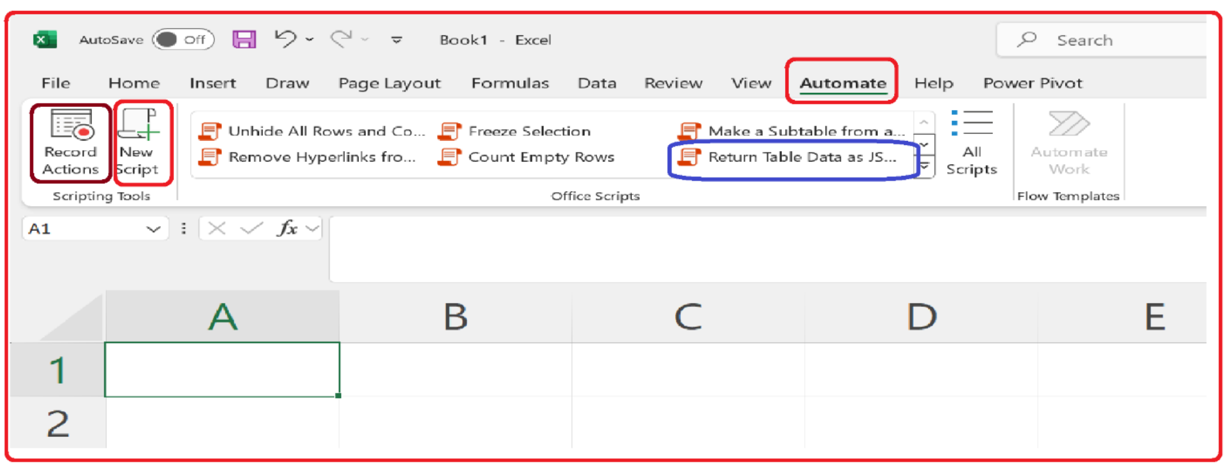
Maybe 2 times a week

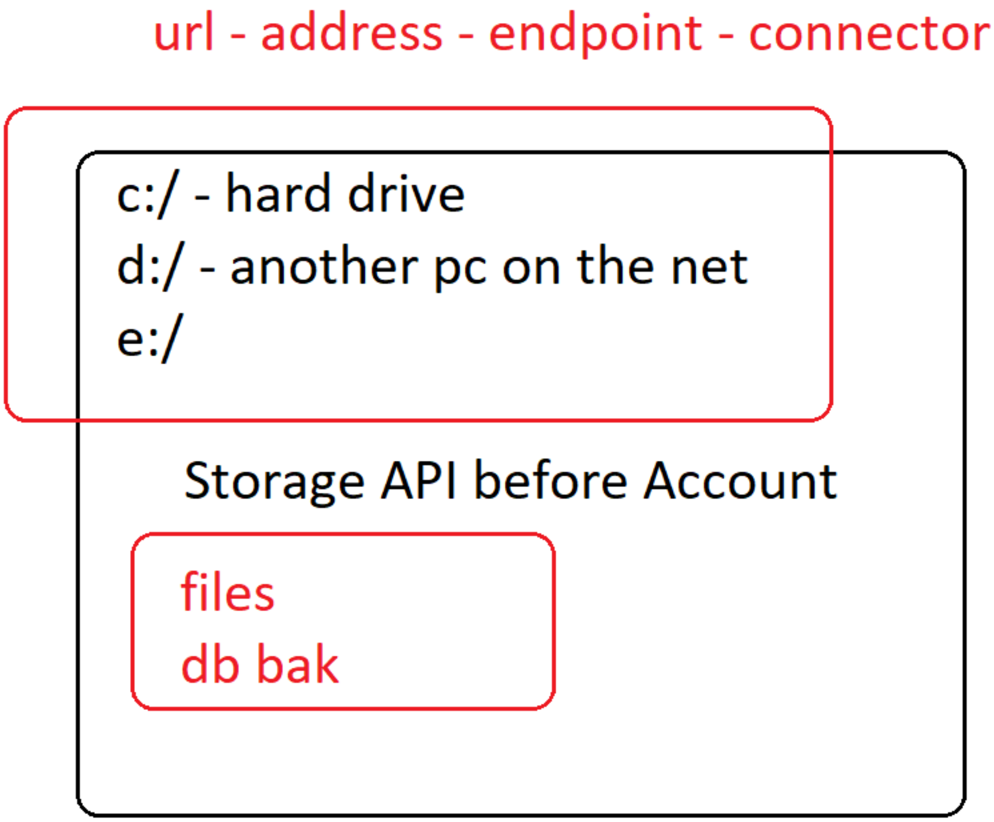
We have rule that says

Data loss can only be 4 minutes

Backup the Logs every 4 minutes every day







SQL Provider – Edit Subnet – Azure Portal

In the Azure portal, if the "SQL provider list" is blank when editing a VLAN subnet, there could be several reasons for this issue. Here are some possible causes:

1. **Missing SQL Provider Registration**:
   - Azure SQL services are part of a specific resource provider (Microsoft.Sql). If the resource provider isn't registered within your subscription, the portal may not display SQL options in the VLAN subnet settings.
   - You can check and register the provider by navigating to **Subscriptions** > **Resource providers** in the Azure portal and ensuring Microsoft.Sql is registered. If it's not, click "Register."
2. **Network Policies or Restrictions**:
   - Ensure there are no specific network policies, firewall rules, or restrictions preventing the SQL service from being available or visible in the subnet you're working on.
   - SQL-based services may not show up if the network infrastructure (such as NSG rules or route tables) is misconfigured or blocking traffic to the SQL endpoints.
3. **VNet Peering/Region Issues**:

   - If your virtual network (VNet) is peered with another VNet, ensure that both VNets are in regions where SQL services are supported. Some regions might have limited SQL services or configurations.
   - Cross-region peering or misconfigured VNets might also cause the SQL options to appear blank.

4. **Service Endpoints Configuration**:
   - To access Azure SQL Database or Managed Instance from a VNet, a service endpoint for Microsoft.Sql must be enabled for the subnet in question.
   - If the service endpoint is not configured correctly, you might see the SQL provider list as empty. Make sure that you've added the correct service endpoint for Azure SQL in your subnet settings.

5. **SQL Service Availability in the Region**:
   - Double-check that the SQL services you are trying to configure are available in the region where the VNet/subnet exists. Occasionally, certain services are region-specific.

6. **Portal or Browser Cache Issues**:
   - Sometimes, the Azure portal might not load certain elements properly due to caching or temporary UI issues. Try refreshing the portal, clearing your browser cache, or using a different browser to see if the problem persists.

7. **Permissions**:
   - Ensure that the account you are using has sufficient permissions to modify the subnet and the associated resources, such as Azure SQL. Lack of permissions may prevent the SQL options from showing.

Actions to Troubleshoot:

- **Check Resource Provider**: Register Microsoft.Sql if it's not already registered.
- **Review Network Configurations**: Ensure correct service endpoints and network policies are in place.
- **Check Permissions**: Verify your role and access level.
- **Inspect the Region**: Ensure SQL services are available in the region.

If none of these steps resolves the issue, you may want to check the activity logs or contact Azure support for further investigation.